Analysis of Composite Kiteboard Design

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Abstract

This research paper presents a comprehensive finite element analysis (FEA) of a kiteboard using ANSYS ACP-Pre to evaluate and optimize its structural performance. The study leverages advanced composite materials, specifically carbon epoxy laminates and a polyurethane core, to enhance key performance metrics such as strength, stiffness, and buoyancy. Kiteboards, as critical equipment in the sport of kiteboarding, must balance lightweight construction with high durability to withstand dynamic loads, impact forces, and environmental stresses. The research begins with utilizing a basic geometry of a kiteboard in ANSYS, since focus has been more on composite analysis. After geometry, the material properties of the core and laminates were defined. A detailed meshing process and application of boundary conditions, including fixed supports and remote displacements, replicate real-world operational scenarios. Key focus areas include the influence of ply orientations and laminate stacking sequences on the board's overall mechanical behaviour. This work contributes to the growing body of knowledge on composite applications in sports equipment and underscores the critical role of simulation in achieving lightweight, high-performance designs tailored to specific use cases. The findings pave the way for future innovations in kiteboard manufacturing, offering insights into material selection and composite analysis.

***Keywords**: Kiteboard, Finite Element Analysis, Composite Materials, ANSYS ACP-Pre, Static Structural*

INTRODUCTION

Kiteboarding, also known as kitesurfing, is a high-adrenaline water sport that combines elements of surfing, wakeboarding, and paragliding. In this dynamic sport, riders use a large, controllable kite to harness wind energy, propelling them across water surfaces while performing jumps, tricks, and glides. This sport relies on the seamless integration of skill, balance, and equipment performance. At the heart of the activity is the kiteboard—a lightweight yet robust board designed to withstand the dynamic forces generated by wind, water, and rider movements. Modern kiteboards, often classified as composite sandwich structures, use advanced materials to balance strength, flexibility, and lightweight properties critical for high performance.

Composite materials[1][2] are a cornerstone of modern engineering, combining two or more distinct components to produce materials with superior and customizable properties. In kiteboard construction, typical composites consist of reinforcement materials, such as carbon or glass fibres, embedded in a resin matrix. This combination offers a high strength-to-weight ratios, resistance to the environmental degradation, and high mechanical versatility. Historically, composites have been applied across a wide range of fields, including aerospace, automotive, and marine industries. Their adaptability makes them particularly well-suited for applications demanding a balance between durability and lightweight design, such as high-performance sports equipment.

An inspiring application of composite technology can be seen in the America's Cup [3] yacht, where Team New Zealand leveraged advanced simulation tools like ANSYS to optimize the vessel's design. The integration of composites allowed for significant reductions in weight, improvements in structural integrity, and increased overall speed and efficiency. This achievement highlighted the potential of composite materials in achieving groundbreaking innovations in performance-critical designs. Inspired by this precedent, the aim was to apply similar principles during the analysis of a kiteboard, exploring how advanced simulation tools and material

optimization could enhance its structural performance.

This research delves into the finite element analysis of a kiteboard using ANSYS ACP, focusing on the modelling and analysis of its composite structure. Key areas of investigation include defining the geometry of the board, selecting appropriate materials and ply orientations, and simulating real-world load scenarios to evaluate stress distributions and deformation. The findings contribute to understanding how composite materials and advanced modelling techniques can revolutionize the design of kiteboards, offering insights into material selection, structural efficiency, and overall performance optimization.



Figure 1: Yacht of Team New Zealand

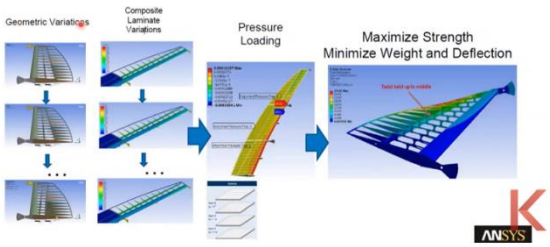


Figure 2: Procedure of Optimization of Yacht Composite Materials

LITERATURE REVIEW

This section outlines the theoretical principles behind key concepts used in composite analysis, including stacking sequences, ply orientations, rosettes, and orientation sequences. Also, it summarizes a research paper which studied composite material analysis on a kiteboard.

The stacking sequence refers to the arrangement of composite plies (layers) in a laminate. Each ply is characterized by its material composition and orientation angle. The sequence in which these plies are stacked determines the laminate's overall mechanical properties, such as stiffness, strength, and resistance to deformation. For example, stacking plies with varying angles (e.g., 0°, 45°, 90°) can enhance a structure's ability to resist multidirectional stresses. This arrangement balances the distribution of loads and minimizes weaknesses in any specific direction, crucial for components like kiteboards that experience complex forces. Ply orientation specifies the angle at which the fibres within a composite ply are aligned relative to a reference axis. Common orientations include 0° (aligned with the principal axis), 45° (diagonal alignment), and 90° (perpendicular to the principal axis). The orientation angle plays a critical role in how the laminate responds to loads:

0° Ply: Maximizes strength and stiffness along the principal axis.

45° Ply: Provides shear resistance and contributes to torsional stiffness.

90° Ply: Enhances strength perpendicular to the primary load direction.

By strategically combining ply orientations, engineers can tailor the laminate to meet specific structural demands.

A rosette is a configuration of multiple strain gauges arranged to measure strains in different directions. While primarily a testing method, the theoretical principle of strain measurement is integral to understanding how stresses are distributed within a composite material. Rosettes provide data that can be used to validate simulation results, ensuring that the modelled stresses align with real-world performance. This is particularly important for complex structures like kiteboards, where stresses vary across the surface due to dynamic forces.

The orientation sequence describes the specific order in which ply orientations are arranged within the laminate. This sequence influences the anisotropic (direction-dependent)

properties of the composite. For instance, a symmetric and balanced orientation sequence ensures that the laminate behaves predictably under loading, minimizing warping or asymmetrical deformation. Engineers use tools like ANSYS ACP to model and optimize these sequences, ensuring that the final structure meets all performance criteria.

To set up a composite analysis in ANSYS Composite PrepPost (ACP), a systematic approach is followed, ensuring each step aligns with the principles of composite material behaviour and structural requirements. The process begins with defining the material properties in the Engineering Data module. Composite materials are anisotropic, meaning their properties vary depending on the direction of the load. This requires the definition of unique parameters such as density, Young's modulus, Poisson's ratio, and shear modulus for both the matrix (e.g., epoxy resin) and reinforcement fibres (e.g., carbon or glass). Directional properties are crucial, as they influence how the material will respond to different types of loads.

Once the material is defined, the next step is to import the geometry of the structure to be analysed. In this case, the kiteboard geometry is imported into the ACP workspace in formats like IGES or STEP. Proper preparation of the geometry ensures that the model is clean, with no open edges or overlapping surfaces that could lead to simulation errors. If needed, pre-processing tools such as ANSYS DesignModeler or SpaceClaim can be used to rectify any issues in the geometry. After geometry import, the meshing process begins. Meshing divides the structure into smaller finite elements, which serve as the foundation for simulation. For composites, the mesh must be refined in areas expected to experience high stresses, as this ensures accurate results. Additionally, the selection of appropriate element types—shell or solid—depends on the laminate structure. Shell elements are often preferred for thin composite layers like those used in kiteboards.

The core of composite analysis lies in defining the ply layup. This step involves specifying the number of plies (layers) in the laminate and their respective orientations. Each ply is assigned a fibre orientation angle, such as 0° for alignment with the principal axis, ±45° for shear resistance, or 90° for strength perpendicular to the primary load direction. This orientation influences the laminate's ability to distribute stresses. The stacking sequence, or the order of these orientations, plays a pivotal role in determining the overall behaviour of the composite. Balanced and symmetric stacking sequences are often used to prevent warping and to ensure consistent mechanical properties in all directions.

In sandwich structures like kiteboards, the core material is specified to provide lightweight buoyancy and energy absorption. The thickness of the core and its interaction with the composite facesheets is critical for achieving optimal bending resistance. Boundary conditions are then applied to replicate real-world scenarios. Fixed supports are defined at regions where the structure will not move, such as edges or mounting points. Loading conditions, such as pressures, forces, or displacements, are applied to simulate the forces acting on the kiteboard during use. This step ensures that the simulation accurately reflects operational conditions. A key feature in ACP is the visualization of the ply layup. This allows engineers to inspect the stacking sequence and thickness distribution in the laminate. Tools like rosettes, which are local coordinate systems for each ply, help track directional stresses and strains, ensuring that the ply orientations align with design goals.
Finally, the setup is validated to ensure consistency in the defined materials, orientations, and boundary conditions. The model is then exported to ANSYS Mechanical for further analysis, such as static structural or dynamic simulations. This step provides detailed insights into stress distribution, deformation, and overall structural integrity. By following this workflow, ANSYS ACP

enables precise modelling and analysis of composite materials, ensuring their optimal performance under complex loading conditions. For the kiteboard analysis, these steps allow the study of its structural behaviour, providing valuable insights for performance optimization.

The study by Stier[4] et al. introduces an innovative approach to designing and optimizing composite kiteboards, aiming for enhanced performance and structural efficiency. The research focuses on a novel foam-core sandwich structure with carbon/epoxy composite facesheets, optimized for weight and durability. The authors utilized advanced computational and experimental techniques, including finite element modelling (FEM) and HyperSizer software, to achieve a lightweight yet robust design tailored to the demanding conditions of kiteboarding. The design features curved edges and a central keel, which replace traditional fins to enhance hydrodynamic stability and reduce localized stress. This geometry improves upwind performance and provides superior gliding characteristics while minimizing manufacturing challenges.

The kiteboard's structural optimization employed a CAD model and extensive material testing, such as tensile and shear tests of carbon fibre composites, ensuring accurate simulation inputs. Two finite element models, one using shell elements and another using solid elements, validated the experimental findings under three-point bending tests. Full-field strain measurements revealed localized deformation patterns, which aligned closely with FEM predictions, showcasing the model's reliability in capturing complex loading scenarios. The final design achieved a 5.7% weight reduction compared to the prototype while adhering to stringent failure criteria, such as core shear crimping and facesheet wrinkling. Additionally, the design promotes manufacturability by employing continuous ply layups across solid and sandwich regions, ensuring practical production.

This research exemplifies the integration of advanced composites, computational tools, and experimental validation to address real-world engineering challenges. The resulting kiteboard combines high performance, safety, and manufacturability, reflecting a significant advancement in sports engineering.

METHODOLOGY

The kiteboard's geometry plays a pivotal role in its performance, particularly in high-intensity water sports such as kiteboarding. Its design reflects a careful balance between streamlined aesthetics and functional efficiency. This is achieved by integrating a wider midsection, strategically tapered ends, and smooth transitional zones, all of which collectively contribute to optimizing buoyancy, speed, manuverability, and durability. The midsection, designated as region A, is the widest part of the board and serves as the primary area for load distribution. This wider profile provides significant buoyancy and stability, ensuring the board remains afloat under the rider's weight while maintaining a steady platform for effective control. The midsection’s design not only supports the rider but also minimizes the risk of the board sinking during use, especially under dynamic loads. At the same time, the board’s tapered ends, referred to as regions B and C, minimize hydrodynamic drag by allowing water to flow smoothly around the board during high-speed motion. These regions are particularly critical for enabling sharp directional changes and enhancing the efficiency of turning manoeuvres, both of which are crucial in the context of kiteboarding. The gradual transitions between the wide midsection and the tapered ends, represented by regions D and E, are specifically engineered to ensure a smooth stress distribution. These transition zones prevent the occurrence of stress concentrations that could compromise the board’s structural integrity. Furthermore, the front (C) and leading edge (F) are optimized to withstand forward and upward forces generated by the kite’s pull and the hydrodynamic flow, while the back (G)

accommodates forces associated with braking and manoeuvring. These carefully designed features work synergistically to deliver a geometry that maximizes dynamic performance, agility, and structural resilience under extreme environmental conditions.

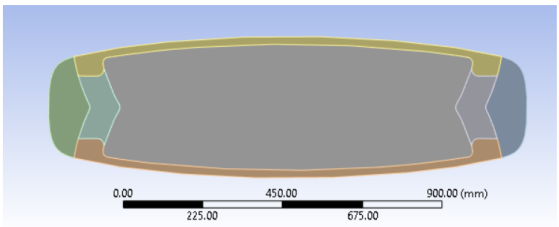


Figure 3: Geometry of Kiteboard[5]

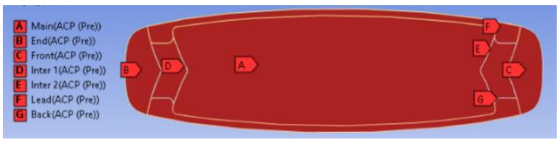


Figure 4: Named Selection on Geometry[5]

The construction of the kiteboard employs an advanced composite material system, integrating a polyurethane sandwich core with carbon-epoxy laminate layers. This combination leverages the complementary properties of both materials to meet the demanding operational requirements of the sport. The polyurethane core serves as the lightweight foundation of the board, chosen for its exceptional compressive strength and buoyancy. Its inherent ability to absorb vibrations enhances the comfort of the ride, particularly under dynamic conditions.

The core's thickness varies across different regions of the board, with values of 28.5 mm, 15.2 mm, and 6.1 mm, arranged in a sandwich configuration along two orthogonal directions. The thicker sections, primarily in the central region of the board, provide structural rigidity and support, bearing the majority of the rider's load. In contrast, the thinner sections, located near the ends of the board, contribute to its flexibility. This flexibility is vital for enabling controlled deformation during sharp manoeuvres, thereby improving the board's responsiveness. The sandwich structure itself is a key innovation, enhancing the stiffness-to-weight ratio by combining the lightweight polyurethane core with stiff outer laminates.

While the core provides a lightweight yet strong base, the outer laminates are designed to handle bending and torsional stresses effectively.

	Property	Value	Unit
1	Property	Value	Unit
2	Material Field Variables	Table	
3	Orthotropic Elasticity		
4	Young's Modulus X direction	60	MPa
5	Young's Modulus Y direction	60	MPa
6	Young's Modulus Z direction	60	MPa
7	Poisson's Ratio XY	0.35	
8	Poisson's Ratio YZ	0.35	
9	Poisson's Ratio XZ	0.35	
10	Shear Modulus XY	23	MPa
11	Shear Modulus YZ	23	MPa
12	Shear Modulus XZ	23	MPa
13	Orthotropic Stress Limits		
14	Tensile X direction	0	MPa
15	Tensile Y direction	0	MPa
16	Tensile Z direction	1.1	MPa
17	Compressive X direction	0	MPa
18	Compressive Y direction	0	MPa
19	Compressive Z direction	0	MPa
20	Shear XY	0	MPa
21	Shear YZ	0.8	MPa
22	Shear XZ	0.8	MPa
23	Ply Type		
24	Type	Orthotro...	

Figure 5: Material Properties of Polyurethane

The outer layers of the kiteboard consist of carbon-epoxy laminates, which were selected for their superior mechanical properties, including high tensile strength, stiffness, and resistance to fatigue. The carbon fibres are particularly effective at managing the tensile and compressive forces exerted on the board during high-speed motion and dynamic manoeuvres. The epoxy resin, serving as the matrix material, ensures strong adhesion between the fibres while providing additional protection against environmental factors such as water exposure and UV degradation.

Each carbon-epoxy layer is 0.2 mm thick and is symmetrically placed above and below the polyurethane core. This symmetric placement enhances the board's resistance to flexural and torsional stresses, ensuring balanced performance during operation. The stacking sequence of the laminates follows a carefully designed orientation: 0°, ±15°, 45°, and 90°.

	A	B	C	D	E
1	Property	Value	Unit		
2	Material Field Variables	Table			
3	Orthotropic Elasticity				
4	Young's Modulus X direction	1.2E+05	MPa		
5	Young's Modulus Y direction	86000	MPa		
6	Young's Modulus Z direction	86000	MPa		
7	Poisson's Ratio XY	0.28			
8	Poisson's Ratio YZ	0.4			
9	Poisson's Ratio XZ	0.28			
10	Shear Modulus XY	47000	MPa		
11	Shear Modulus YZ	31000	MPa		
12	Shear Modulus XZ	47000	MPa		
13	Orthotropic Stress Limits				
14	Tensile X direction	2274	MPa		
15	Tensile Y direction	30	MPa		
16	Tensile Z direction	30	MPa		
17	Compressive X direction	-1100	MPa		
18	Compressive Y direction	-100	MPa		
19	Compressive Z direction	-100	MPa		
20	Shear XY	60	MPa		
21	Shear YZ	30	MPa		
22	Shear XZ	60	MPa		
23	Orthotropic Strain Limits				
24	Tensile X direction	0.017			
25	Tensile Y direction	0.003			
26	Tensile Z direction	0.003			
27	Compressive X direction	-0.011			
28	Compressive Y direction	-0.019			
29	Compressive Z direction	-0.019			
30	Shear XY	0.01			
31	Shear YZ	0			
32	Shear XZ	0			
33	Puck Constants				
34	Material Classification	Carbon			
35	Compressive Inclination XZ	0.3			
36	Compressive Inclination YZ	0.25			
37	Tensile Inclination XZ	0.35			
38	Tensile Inclination YZ	0.25			

Figure 6: Material Properties of Carbon Epoxy

The 0° layers align with the board's longitudinal axis, providing maximum strength against bending forces. The ±15° layers enhance resistance to shear forces and mitigate the risk of delamination under torsional loads. The 45° layers contribute to multi-directional stiffness, improving torsional rigidity, while the 90° layers provide transverse strength to withstand distributed loads. This orientation, combined with the symmetric stacking, ensures consistent stiffness distribution across the board, preventing warping or twisting during use and enhancing overall durability.

The finite element model (FEM) of the kiteboard is developed to simulate its structural behaviour under real-world operational loads. The computational model begins with the creation of a structured quadrilateral (quad) mesh, which discretizes the kiteboard's geometry into smaller, manageable elements for numerical analysis. Quadrilateral elements are specifically chosen for their superior performance in bending-dominated problems, making them particularly suitable for plate-like structures such as kiteboards. The mesh design incorporates refinement in regions anticipated to experience higher stress or strain, such as areas near the fixed support and those subjected to remote displacements. This ensures that critical zones are accurately represented, capturing stress concentrations and localized deformations effectively. Conversely, the central regions of the board, which are less critical to the analysis, are assigned a coarser mesh to reduce computational demand. This structured approach to meshing ensures alignment with the geometry, minimizes distortion, and enhances the reliability of the simulation results. By balancing computational efficiency with numerical accuracy, the mesh design allows the FEM to yield robust and representative outcomes.

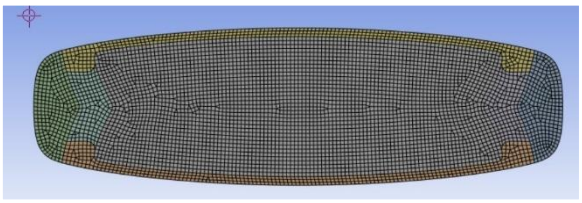


Figure 7: *Mesh of the Geometry*[5]

Boundary conditions are essential for replicating the operational environment of the kiteboard within the simulation. A fixed support boundary condition is applied to the left end of the board, fully constraining all degrees of freedom—translations and rotations—in that region. This fixed support anchors the kiteboard, providing a stable reference point for deformation calculations. On the right end, a remote displacement boundary condition is applied to simulate controlled movements and rotations. The remote displacement imposes a vertical translation of 2 mm in the Y direction, mimicking the bending effect caused by hydrodynamic pressure and the rider's weight. Additionally, a 5° rotation about the X-axis represents torsional loading due to uneven forces, while a 2° rotation about the Y-axis captures lateral twisting effects induced by wave dynamics or wind pressure. These specific displacements and rotations were selected to reflect critical loading scenarios

encountered during kiteboarding. To prevent over-constraining the model and inducing artificial stress concentrations, the X translation and Z-axis rotation are left unconstrained. This allows the board to respond naturally to external forces, ensuring that the simulation accurately captures its mechanical behaviour.

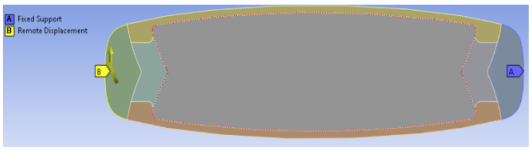


Figure 6: Fixed Support and Remote Displacement[5]

The displacement and rotation values used in the model were meticulously chosen to align with the most critical loading scenarios likely to occur during kiteboarding. A vertical displacement of 2 mm in the Y direction is applied, simulating the bending effect resulting from the rider's weight and hydrodynamic forces acting on the board's surface. This displacement replicates the primary mode of deformation experienced by the board during typical operation. Additionally, a 5° rotation about the X-axis represents torsional loading caused by asymmetric forces, such as uneven pressure applied by the rider's feet during manoeuvres. This torsional rotation is crucial for evaluating the kiteboard's ability to withstand twisting motions, which are common during high-performance activities such as sharp turns or wave riding. Furthermore, a 2° rotation about the Y-axis is included to capture transverse twisting effects caused by lateral forces. These forces may arise due to uneven water pressure during turning manoeuvres or dynamic interactions with wind and waves. The inclusion of these displacements and rotations ensures the model represents the primary deformation modes critical to understanding the kiteboard's performance under operational loads.

To prevent over-constraining the simulation, certain degrees of freedom are left unrestrained. For instance, the X translation and Z-axis rotation remain unconstrained, allowing the board to move or rotate freely in these directions. These degrees of freedom were determined to have a minimal impact on the board's structural behaviour under the selected loading conditions. By avoiding unnecessary constraints, the model ensures that artificial stress concentrations or unrealistic restrictions are not introduced, thereby preserving the physical authenticity of the simulation.

The deformable nature of the remote displacement configuration allows the kiteboard to exhibit realistic flexural and torsional behaviour under applied loads. This is critical for evaluating the board's performance in scenarios involving bending, twisting, and other complex loading conditions. The combination of bending deformation in the Y direction, torsional rotation about the X-axis, and lateral twisting along the Y-axis provides a comprehensive understanding of the board's mechanical responses to real-world forces. These specific loading conditions were prioritized because they represent the primary operational stresses encountered during kiteboarding, ensuring that the finite element analysis delivers relevant and actionable insights.

In conclusion, the finite element model of the kiteboard integrates a well-structured approach to boundary conditions and displacement configurations, ensuring a highly accurate representation of its structural behaviour. The use of fixed supports at one end and remote displacement at the other mimics the real-world constraints and forces acting on the board, capturing its dynamic response under critical loading scenarios. The careful selection of displacements and rotations emphasizes bending and torsional effects, which are central to the kiteboard's performance and structural integrity. This model not only provides a detailed assessment of the stress, strain, and deformation characteristics of the board but also establishes a robust foundation for subsequent design optimization and performance evaluation.

RESULTS AND DISCUSSIONS

The results of the structural analysis conducted on the composite kiteboard using finite element analysis (FEA) provide key insights into its mechanical behaviour under typical operational loads. Two primary parameters—equivalent stress distribution and total deformation—were studied in detail to evaluate the performance of the kiteboard's composite structure in dynamic environments.

The analysis of equivalent (von Mises) stress distribution reveals that the maximum stress is concentrated near the edges of the kiteboard, particularly close to the mounting points. These areas experience the highest loads due to the combined forces exerted by the rider's weight and the kite's pull. The maximum stress value was found to be approximately 603.9 MPa. The outer carbon fibre layers, which serve as the primary load-bearing elements of the board, effectively handle these tensile and compressive stresses, ensuring the overall structural stability. Across the rest of the board, the stress levels are significantly lower and exhibit a uniform distribution. This uniformity is a result of the composite construction, which reduces the likelihood of localized material failure and enhances the board's durability under varying conditions. However, the high-stress regions near the mounting points suggest the potential need for reinforcement. To address this, the addition of hybrid layers or thicker laminates in these critical zones could further bolster the board's ability to withstand concentrated stresses, improving both performance and longevity. Despite the observed stress concentrations, the analysis confirms that the materials remain well within their permissible strength limits, underscoring their suitability for high-performance kiteboarding applications.

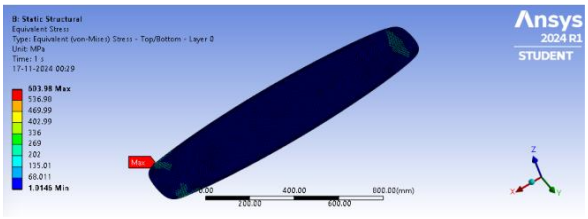


Figure 9: von Mises Stress Contour[5]

The deformation analysis complements the stress findings, providing a detailed view of how the kiteboard responds to applied forces. The maximum deformation was observed at the centre of the board, reaching a value of 14.012 mm, while the edges and reinforcement zones exhibited minimal displacement. This deformation pattern reflects the balance between rigidity and flexibility that the composite design seeks to achieve. The carbon fibre layers provide the necessary stiffness to preserve the board's structural integrity under load, while the inherent flexibility of the composite materials allows for sufficient deformation to enhance manuverability. This combination ensures the kiteboard can adapt to the dynamic forces encountered during use, such as rapid directional changes and varying hydrodynamic pressures.

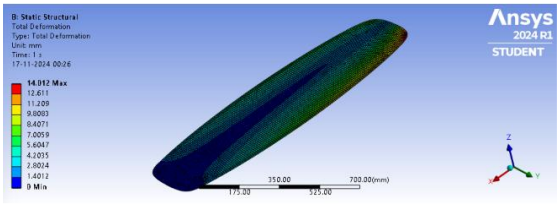


Figure 10: Displacement Contour[5]

The minimal deformation observed in the edge regions aligns closely with the stress concentration findings, reinforcing the effectiveness of the material configuration in these zones. By maintaining low deformation at the edges while allowing controlled flex in the central portion, the kiteboard achieves both stability and responsiveness. Additionally, the lightweight nature of the polyurethane core and carbon-epoxy laminate construction contributes to the board's overall handling efficiency. The reduced mass enhances the

rider's ability to execute precise movements and sharp turns, a critical requirement for high-performance kiteboarding.

Taken together, the results of the stress and deformation analyses provide a comprehensive understanding of the kiteboard's structural behaviour. The equivalent stress distribution highlights the areas that may require reinforcement, while the deformation analysis demonstrates the board's ability to balance stiffness and flexibility. The composite material system, with its strategically varied thickness and layering, proves highly effective in meeting the demanding conditions of kiteboarding. By ensuring strength, durability, and agility, the kiteboard's design and construction are well-suited to withstand extreme loads and dynamic movements, making it an optimal choice for water sports enthusiasts.

CONCLUSION

The finite element analysis (FEA) conducted on the composite kiteboard has provided significant insights into its structural performance under operational conditions. Utilizing ANSYS ACP for the simulation allowed for an in-depth exploration of the mechanical behaviour of the kiteboard, particularly focusing on stress distribution and deformation. The results highlight the critical role of material properties, ply orientations, stacking sequence, and overall composite layup in determining the performance and durability of the kiteboard.

The equivalent (von-Mises) stress analysis revealed the zones of maximum stress concentration, with values peaking at specific points due to localized loading and boundary conditions. This understanding is crucial for identifying potential failure regions and optimizing the design to ensure safety and reliability during use. The deformation analysis further confirmed the structural integrity of the kiteboard, with controlled displacement within acceptable limits. This validates the efficacy of the composite material selection and layup configuration in withstanding the hydrodynamic and aerodynamic forces encountered during kiteboarding.

Moreover, the study underscores the importance of tailoring the composite layup through the strategic placement of ply orientations. The integration of core materials, such as polyurethane, with carbon fibre/epoxy laminates ensures a balance between weight, stiffness, and strength. The outcomes also emphasize the versatility and accuracy of FEA tools like ANSYS ACP in simulating the performance of complex composite structures, enabling iterative design enhancements with reduced physical prototyping.

The findings from this research pave the way for further exploration in the field of kiteboard design, particularly in optimizing material usage, reducing weight, and improving resilience under extreme conditions. Future work can expand on this study by incorporating dynamic load cases, fatigue analysis, and the effect of varying environmental factors such as temperature and moisture. By leveraging advanced computational tools and materials science, the kiteboarding industry can achieve safer, more efficient, and high-performing boards, enhancing the overall user experience and operational safety.

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